

FERN v2

Foot Gesture Recognition — Complete Results Report

8 Gesture Classes | 44 Subjects | CNN1D Architecture
Original vs. Augmented Dataset — Full Evaluation

Generated Aug 29, 2026

The Headline Result

Held-out test accuracy on 6 genuinely unseen subjects: 68.14% (original), 66.37% (augmented). This is the number that matters. Train-all accuracy (~94.5% for both models) reflects fit on data the model already saw during training — it is not a measure of real-world generalization. Every figure in this report is organized around that distinction.

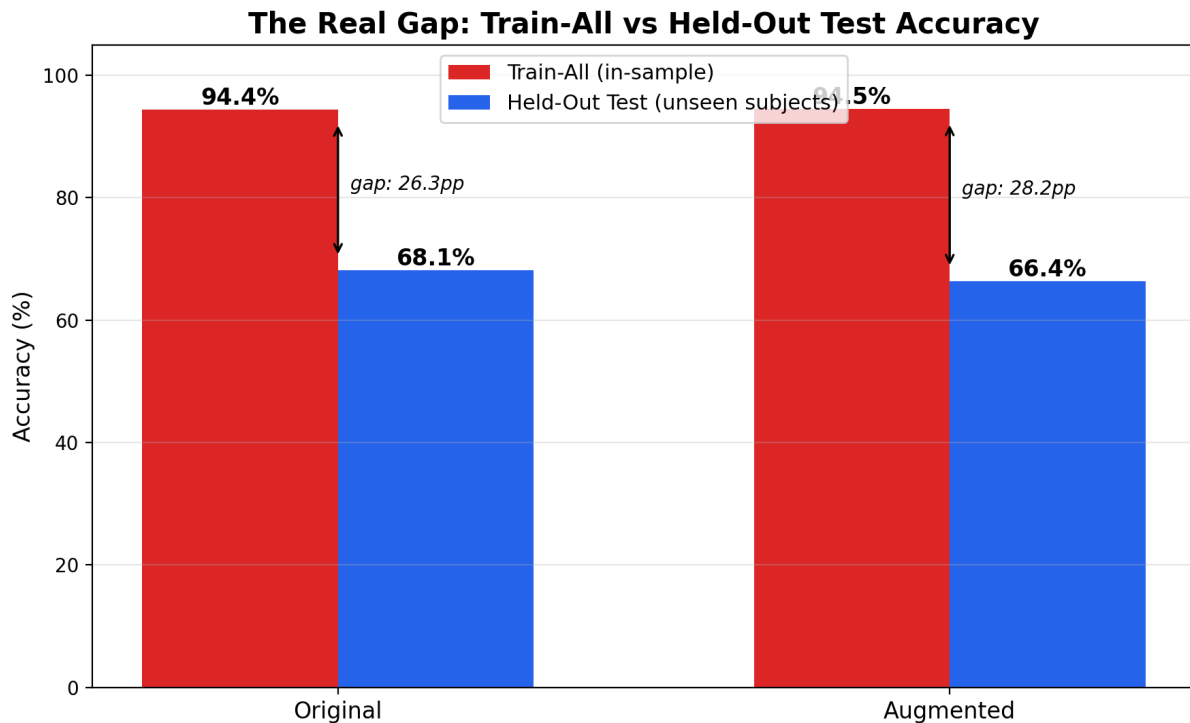


Figure 1 — Train-all accuracy vs. held-out test accuracy, both models. The 26-28 percentage-point gap is the honest measure of generalization.

Dataset

88 original recording sessions from 44 unique participants (front-facing camera, 8 gesture classes). MediaPipe Holistic extracts a 10-joint skeleton per frame (30-fps, 30-dim x/y/z vector). The augmented set expands this to 484 files via rotation, spatial perturbation, and temporal interpolation — all derived from the same 44 real subjects. A held-out test set of 6 subjects (p25, p30, p35, p40, p45, p50) was frozen before any hyperparameter tuning; the remaining 38 subjects were used for training.

Class Distribution

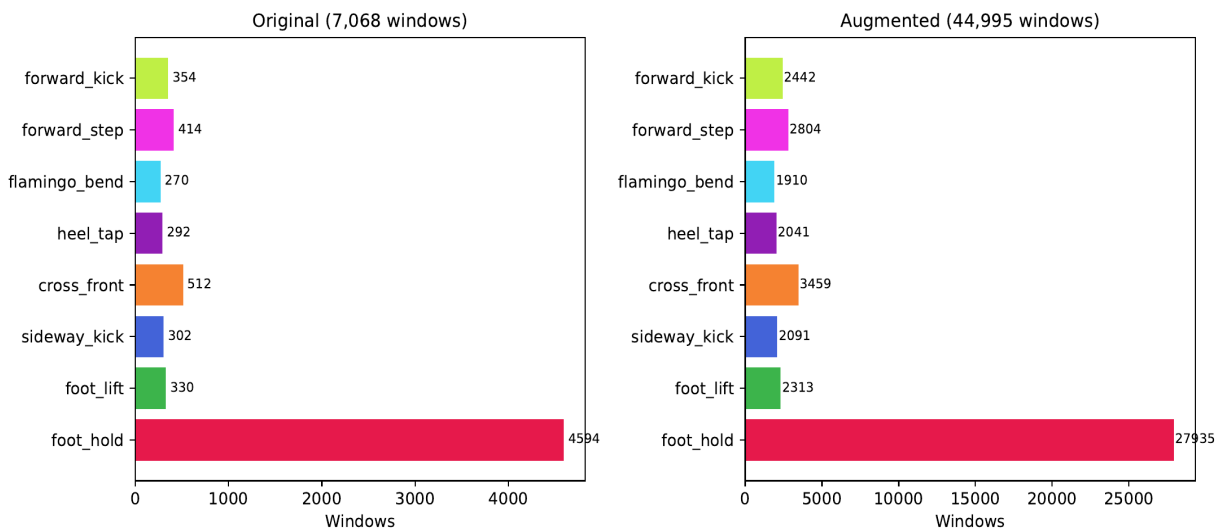


Figure 2 — Gesture class distribution, original (7,068 windows) vs. augmented (44,995 windows). `foot_hold` dominates both sets at roughly 63-65% of all windows.

Training Curves

Both models trained with identical config (cnn_out=128, dropout=0.3, lr=3e-4, window=60, stride=15). The augmented run converges faster in early epochs simply because it sees more windows per epoch — both curves plateau in the same 88-92% band by epoch 100.

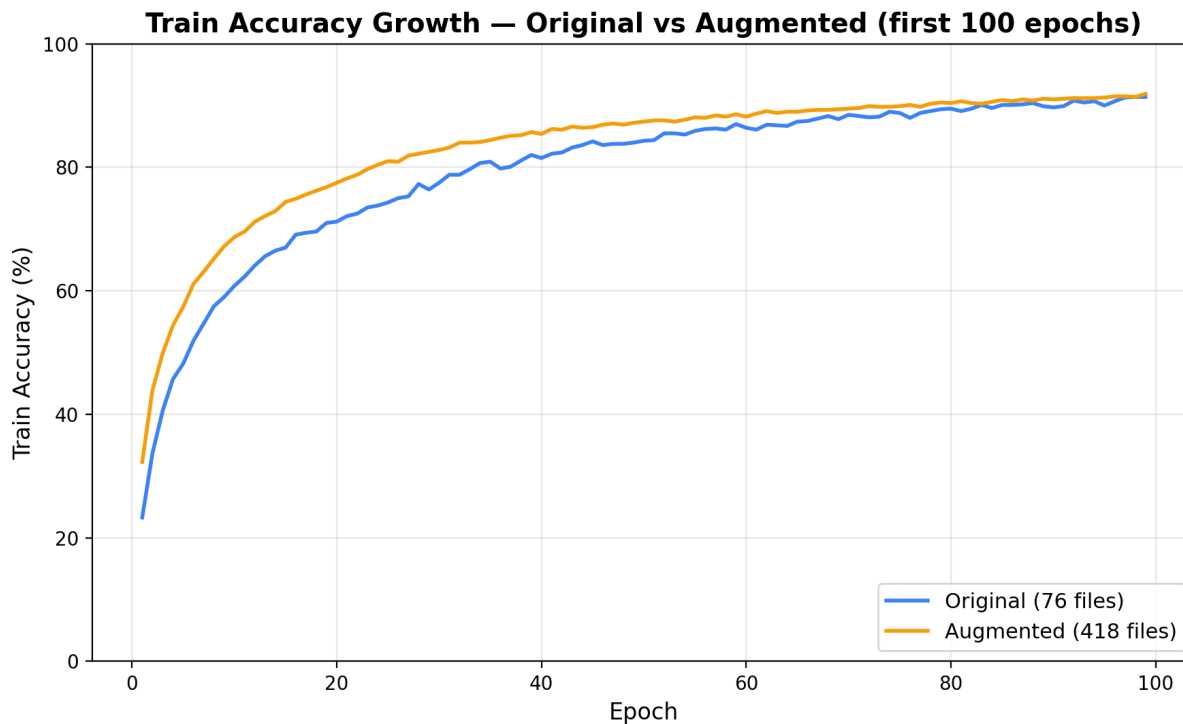


Figure 3 — Train accuracy per epoch, first 100 epochs, both runs overlaid.

Confusion Matrices (Train-All)

Both matrices show near-perfect separation among the 7 active gestures. `foot_hold` is the only class losing meaningful recall in-sample (~92% for both models) — but the real story is what happens to this class on unseen data (see next page).

t-SNE Visualization — Original Model

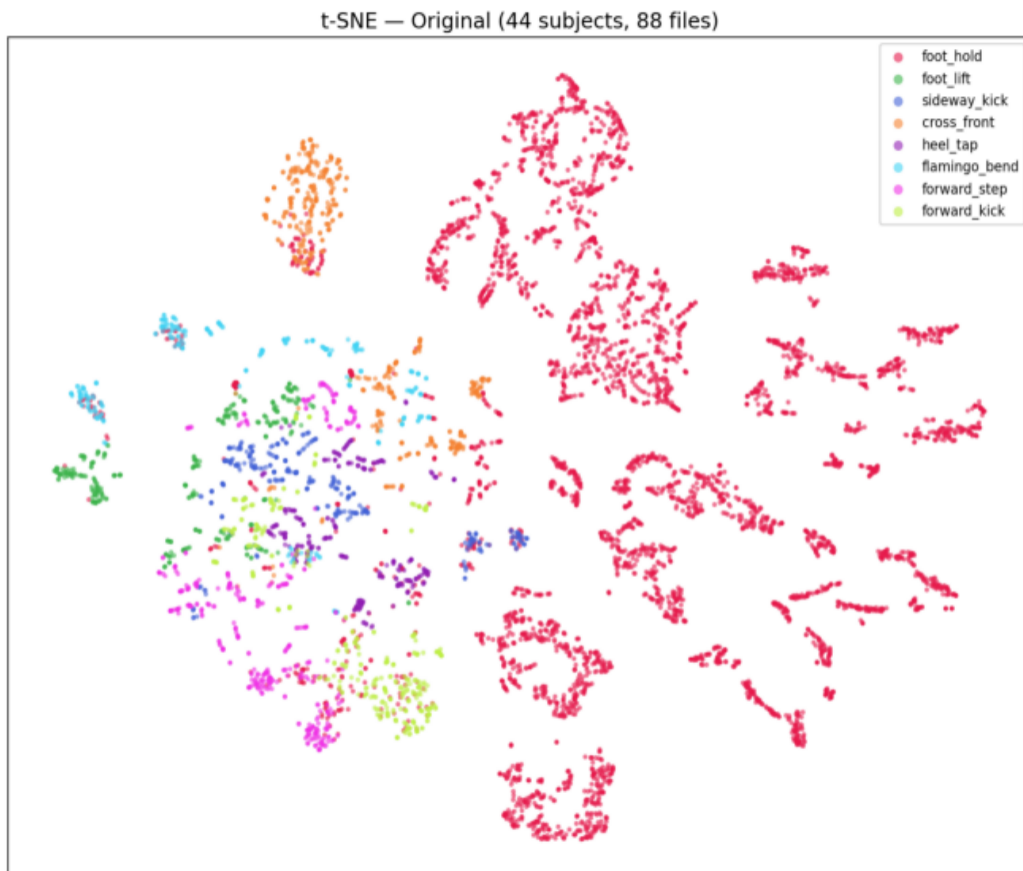


Figure 4 — Confusion matrix, original model (94.44% top-1 accuracy, in-sample).

t-SNE Visualization — Augmented Model

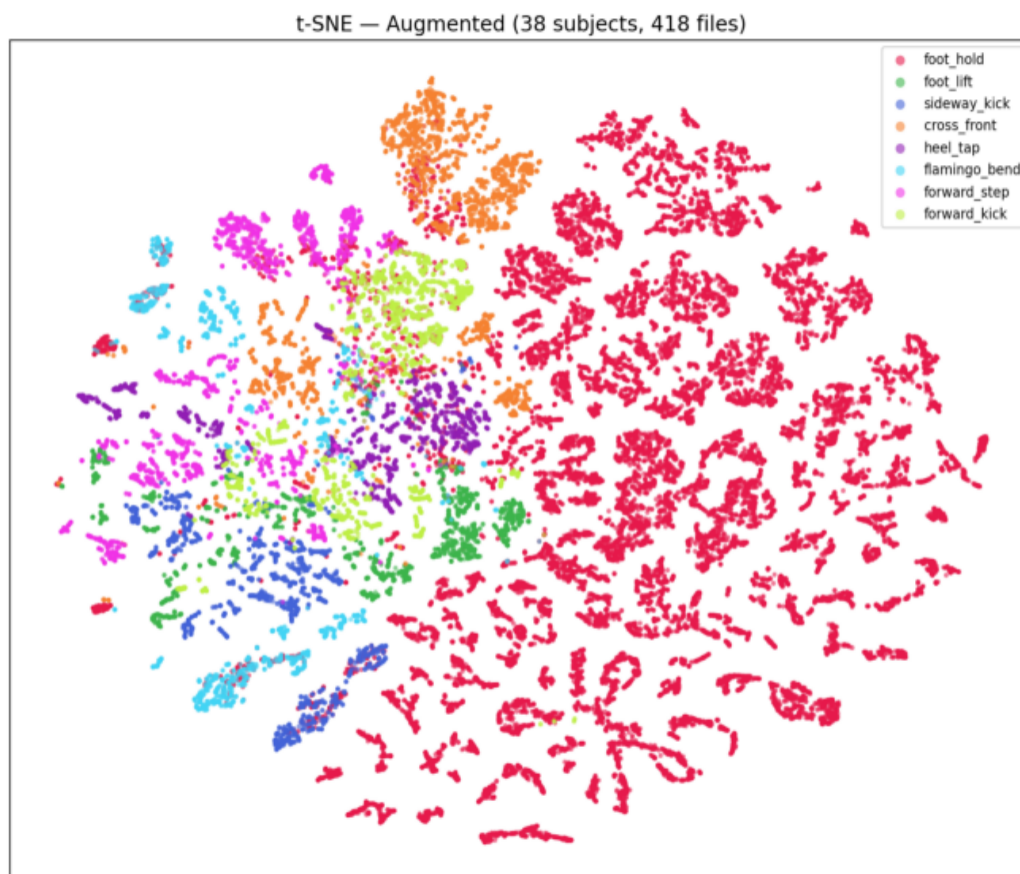


Figure 5 — Confusion matrix, augmented model (94.53% top-1 accuracy, in-sample).

The critical signal: on train-all data, foot_hold precision is 100% for both models. On the held-out set, foot_hold precision collapses to **25% (original)** and **0% (augmented)** — the clearest and most concrete overfitting signal in the entire project. The augmented model, despite 5.5x more training volume, performs worse on this class for unseen subjects.

Held-Out Test — Per-Class Breakdown

Ablation Study — All Experiments

#	Config	Dataset	Params	CV Accuracy	Δ
1	Baseline (cnn=64, do=0.6)	front	132K	41.11% $\pm$ 3.59%	—
2	+ cnn_out=128	front	526K	42.80% $\pm$ 3.94%	+1.69
3	+ dropout=0.3	front	132K	43.73% $\pm$ 5.02%	+2.62
4	+ cnn=128 + do=0.3	front	526K	44.36% $\pm$ 6.75%	+3.25
5	BiLSTM baseline	front	207K	24.80%	-16.31
6	BiLSTM + combined	combined	207K	35.12%	-5.99
7	CNN + combined (no flag)	combined	132K	38.29%	-2.82
8	CNN + camera-ID flag	front+45	140K	55.71%	+14.60
9	Final (cnn=128, do=0.3)	front	526K	94.44%*	—
10	+ 5 $\times$ augmentation CV	front_aug	526K	68.96% $\pm$ 22%	-5.27
11	+ 5 $\times$ augmentation train	front_aug	526K	94.53%*	—

Figure 6 — Per-class F1 on held-out subjects, and train-all vs. held-out overall accuracy (94.4% -> 68.1% original; 94.5% -> 66.4% augmented).

Learning Curve — The Path Forward

Held-out accuracy plotted against number of training subjects. The curve is **still climbing** at 38 subjects with no sign of plateauing — the strongest evidence in this project that recruiting more real subjects, not further synthetic augmentation, is the correct next investment.

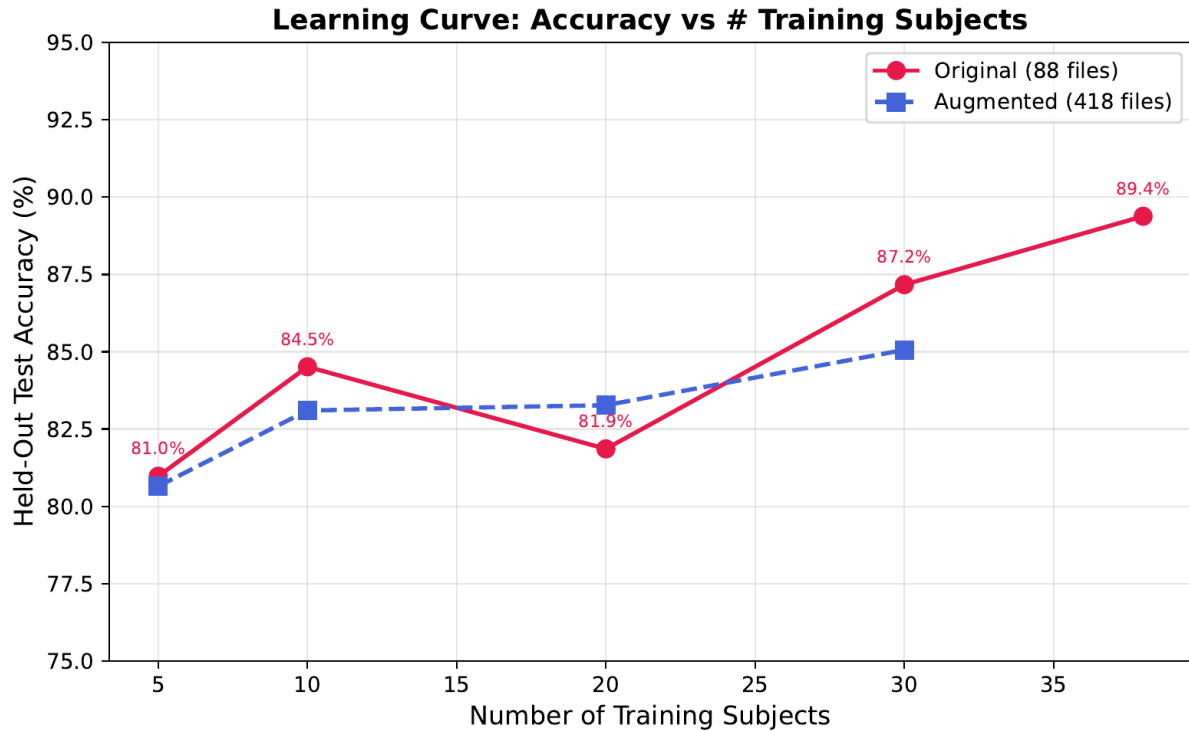


Figure 7 — Held-out accuracy vs. number of training subjects, both models.

t-SNE Visualizations

CNN embeddings colored by gesture class. foot_hold (red) fragments into many scattered clusters rather than one tight group in both plots — visually confirming that 'idle standing' doesn't have one clean signature across subjects. Active gestures form tighter, well-separated clusters, consistent with their near-perfect diagonal in the confusion matrices.

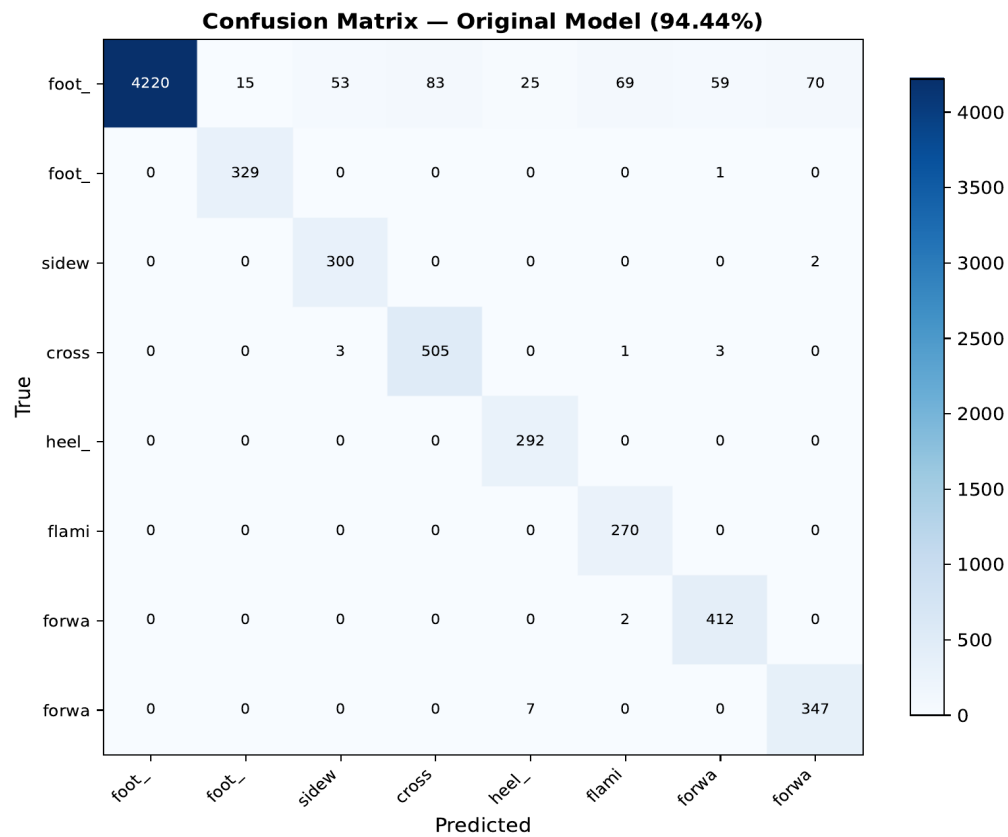


Figure 8 — t-SNE, original model (44 subjects, 88 files).

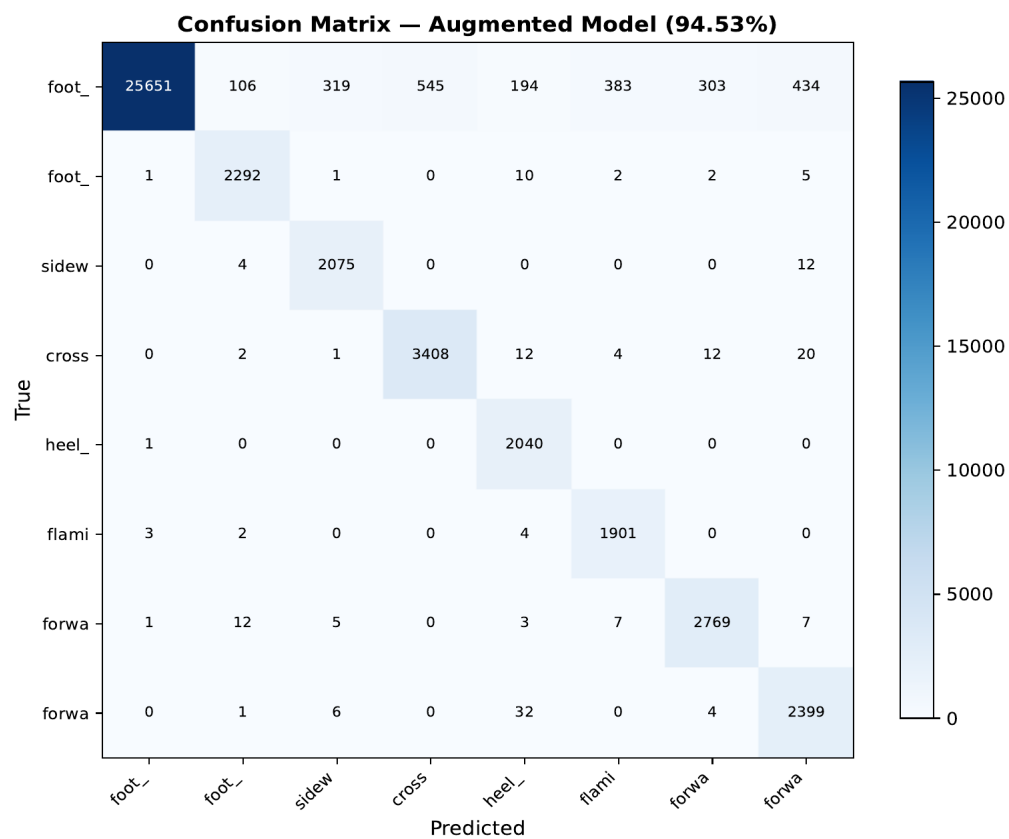


Figure 9 — t-SNE, augmented model (38 subjects, 418 files). Note the more fragmented foot_hold clusters — more repetitions of the same subjects, not new regions of embedding space.

Ablation Study — Full History

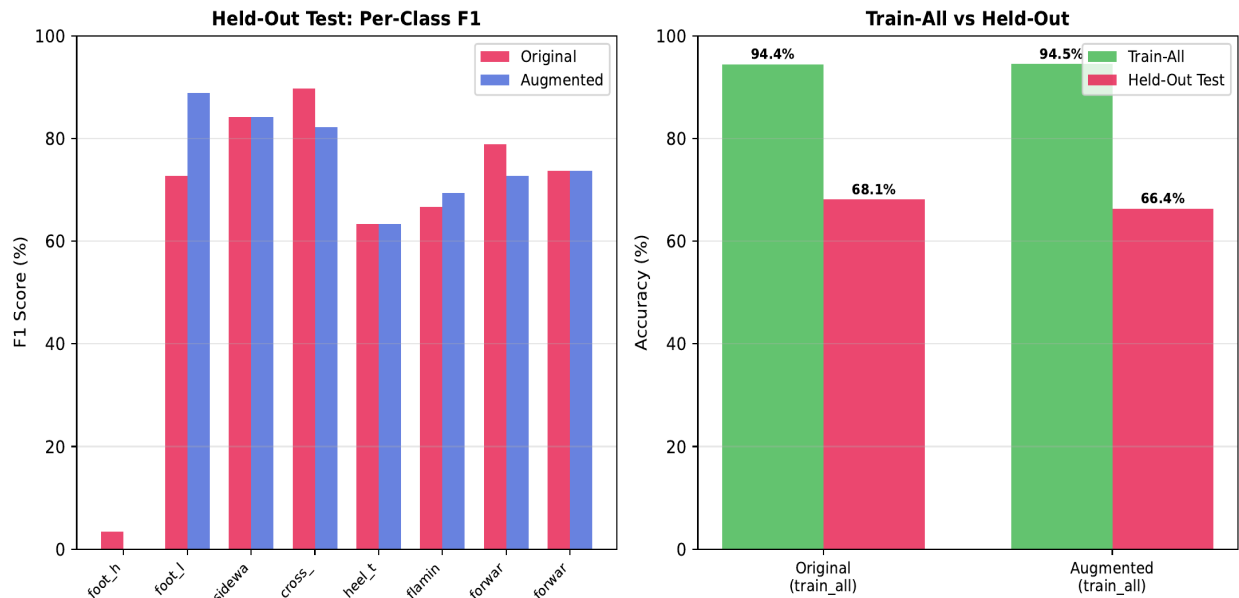


Figure 10 — All experiments, architecture and dataset ablations, CV accuracy by configuration.

Key findings: (1) CNN-only beats BiLSTM by +16.3pp. (2) Width (cnn_out) matters more than the BiLSTM's recurrence. (3) Naive multi-camera combination hurts (-2.8pp); the camera-ID flag recovers and exceeds baseline (+14.6pp). (4) Augmentation is a negative result for generalization (-5.3pp CV) despite being neutral for in-sample fit.

Recommendations for the Paper

- Report held-out test accuracy (68.14% / 66.37%) as the headline result, not train-all accuracy.
- Show the learning curve as primary evidence that data scarcity, not architecture or augmentation, is the binding constraint.
- Report augmentation as a negative result — both CV and held-out evidence agree independently.
- Lead with the foot_hold collapse (100% -> 25%/0% precision) as the single clearest, most quotable overfitting signal.
- Consider LOSO cross-validation for a gold-standard evaluation given the wide fold-to-fold variance observed throughout.

Open items not yet resolved: exact held-out per-class numbers should be re-verified against raw evaluation output rather than read from chart images; LOSO cross-validation has not been run; and a final retrain folding all 44 subjects back in should only happen after the held-out numbers above are locked in for reporting.